

SAM JOSEPH

Robotics Engineer | Embedded Systems Enthusiast | AI & ML Practitioner

B.E. Computer Science (AI & Robotics) | Sathyabama University | 02/2024 – Current

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SUMMARY

Computer Science, Artificial Intelligence, and Robotics student with hands-on experience in embedded systems, robotics, CAD, digital fabrication, sensor integration, and machine learning. Passionate about assistive technology, automation, and real-world robotics applications. Skilled in end-to-end product development from concept, design, and prototyping to testing and validation. Strong problem solver with a keen interest in R&D, mechatronics, UAV systems, and advanced manufacturing.

EXPERIENCE

Research and Development Intern 05/2026 – 07/2026 (3 Months)

Symbiotic Pvt. Ltd., Chennai

- Contributing to the design and development of robotic and assistive-technology product prototypes.
- Conduct in-depth literature and patent reviews to support R&D initiatives.
- Design prototype mechanisms and translate industrial design concepts into functional solutions.
- Conducting tests and evaluations to support R&D decisions and identify relevant technical approaches.
- Collaborate with cross-functional engineering teams from concept, validation, testing, to document prototype development.

Machine Learning Intern 01/2026 – 03/2026 (3 Months)

Edu-versity, Chennai

- Developed machine learning models for educational applications.
- Performed data preprocessing, feature engineering, and exploratory data analysis.
- Used data-driven insights to identify patterns and support practical application development.

EDUCATION

B.E. in Computer Science, Artificial Intelligence & Robotics 02/2024 – Current

Sathyabama University, Chennai, India

- CGPA: 7.74/10 (4th Semester)
- Specialization Focus: AI, Robotics, Embedded Systems
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Embedded Systems, Control Systems, Robotics, Machine Learning
- Active member of Robotics & Innovation Communities
- Hands-on projects in robotics, embedded systems, and automation

ACHIEVEMENTS

- Top 4 in Threx Hackathon 2025 (Team Event)
- Selected for Second Round in Smart India Hackathon (SIH) 2025
- Demonstrated proficiency in coding fundamentals, data structures, and algorithm problem solving.
- Active participant in Robotics & Innovation activities and technical competitions.
- Certificate of Participation – Various workshops on Robotics, IoT & AI.
- Consistent academic performer with strong analytical and problem-solving skills.

PROJECTS

Intelligent Rider Safety & Vehicle Interlock System

Developed an ESP32-based intelligent rider safety system integrating alcohol detection, GPS tracking, GSM emergency alerts, and automatic vehicle ignition lock.

IoT-Based Autonomous Solar Tracking System

Designed and implemented an IoT-enabled dual-axis solar tracking system that optimizes photovoltaic panel orientation using real-time sensor feedback and embedded control.

EMG-Controlled Assistive Robotic Arm

Developed an assistive robotic arm controlled using Electromyography (EMG) signals to enable intuitive human-machine interaction for individuals with limited upper limb mobility. Focused on real-time signal acquisition, processing, and multi-DOF control for smooth motion and adaptive grasping.

Autonomous Precision Agriculture Rover

Designed and developed an autonomous agricultural rover powered by a Raspberry Pi single-board computer for precision farming applications.

Quadruped Robotic Platform (Rex)

Designed and developed a Raspberry Pi-powered quadruped robotic platform focused on real-time locomotion, sensor integration, and embedded control.

Modular Multi-DOF Robotic Manipulator

Designed and developed a multi-DOF robotic manipulator with CAD-designed and 3D printed components, integrating servo-based actuation for precise object manipulation.

TECHNICAL SKILLS

Programming & Software

Python • C • C++ • Embedded C
Microcontroller Programming • Debugging
Git • Linux • VS Code

Robotics & Autonomous Systems

Mobile Robotics • Robotic Manipulators
Kinematics & Dynamics • Path Planning (Basic)
Sensor Fusion • SLAM (Basic)
Control Systems • ROS (Basic)

Embedded Systems & IoT

ESP32 • Arduino • Raspberry Pi
Peripheral Interfacing • RTOS (Basic)
Sensor Integration (IMU, GPS, Ultrasonic, IR, etc.) • Actuator Control
IoT Protocols (MQTT, HTTP)

AI & Computer Vision

OpenCV • Image Processing
Machine Learning • Model Training
Data Analysis • Scikit-learn • NumPy • Pandas

CAD & Mechanical Design

Fusion 360 • SolidWorks • Autodesk Inventor
3D Modeling • Assembly Design
Engineering Drawings • GD&T (Basic)
DFM (Design for Manufacturing)

Manufacturing & Fabrication

FDM 3D Printing • Resin 3D Printing
3D Printer Operation • CNC Machining
VMC Operation • CNC Setup & Basic Tool Handling
Laser Cutting • Sheet Metal Basics
Prototype Fabrication • Mechanical Assembly
Finishing & Post-Processing • Digital Fabrication

UAV & Drone Skills

Drone Piloting • Quadcopter Operations
UAV Flight Planning • Basic UAV Systems
Pre-flight Checks • Mission Planning (Basic)
Flight Safety • Payload Handling (Basic)

R&D & Engineering Practices

Problem Definition • Technical Research
Literature & Patent Review • Concept Development
Rapid Prototyping • Design Iteration
Testing & Validation • Failure Analysis
Root-Cause Analysis • Technical Documentation
Test Reports • Data Logging • Version Control

Testing & Quality

Functional Testing • System Identification
Measurement & Instrumentation (Basic)
Troubleshooting • Debugging
Reliability-Oriented Design

Tools & Platforms

GitHub • MATLAB (Basic) • Arduino IDE
Linux • KiCad (Basic) • Proteus (Basic)

LANGUAGES

• Tamil Native
• English Advanced (C1)
• French Elementary (A2)
• Hindi Elementary (A2)

LINKS [sam-black007](https://sam-black007.github.io)
 [linkedin.com/in/sam-joseph-timothy007](https://www.linkedin.com/in/sam-joseph-timothy007)